USAF ACADEMY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
LAB #1 - Transmission Line Reflection Coefficients
Spring 2025

COLLABORATION POLICY: The lab itself is a team assignment; each team should submit one combined lab report.

AUTHORIZED RESOURCES: Any resources are authorized.

DOCUMENTATION: You must include a documentation statement with each assignment in accordance with the course policy letter.

DUE DATES: The lab report will be due as indicated on Gradescope.

INSTRUCTIONS:

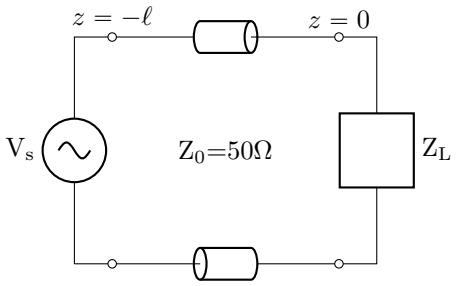
- Before lab: read entire lab packet.
- Before lab: read the VNA user's guide
- During lab: complete the measurements as specified.

EQUIPMENT SAFETY:

- Equipment safety rules must be observed at all times and are for the safety of the students and protection of the equipment.
- No food or drink is allowed in the ECE labs while conducting measurements. Cadets are requested to finish their snacks outside the classroom and place all trash in the trash containers before class starts, and keep all drinks off work surfaces.
- At least two people (instructor/technician/cadet) must be in the room when equipment is in use.
- Cadets should return equipment to original configuration before leaving.
- Cadets must clean up their immediate area (return wires, pick up trash, etc.) before leaving.
- Cadets are to refrain from "playing" with the equipment, parts, and wires when there is no hands-on activity underway.
- When plugging in or unplugging wires, make sure any signal sources (AC or DC) are not transmitting.

1 Purpose

The goal of this lab is to use a Vector Network Analyzer (VNA) to measure the reflection and transmission coefficients in a transmission line system as shown below. You will familiarize yourself with RF measurement techniques, calibrate the VNA, then measure several known load impedances, and, finally, characterize an unknown device.



2 Background

In this lab, we will measure the reflection and transmission coefficient for some devices over various frequency ranges. In these ranges, we can observe reflection and transmission as described in class. Recall the reflection and transmission coefficients are found by:

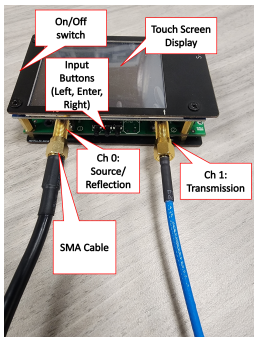
$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$
$$T_L = 1 - \Gamma_L$$

In this lab, we won't be considering the length of our transmission lines (ℓ), because we will calibrate the length out of our calculations.

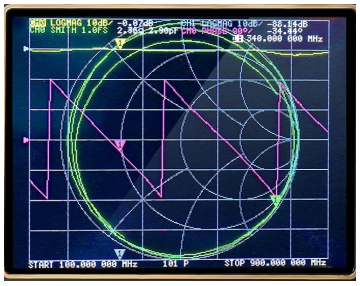
3 Equipment

3.1 Basic Familiarization

A full user's guide for the VNA is available online. Please familiarize yourself with this guide **before** the lab period. The VNA is shown in Figure 1a and some of the key components are highlighted.



(a) An annotated overview of the VNA.



(b) An initial view of the nano VNA screen.

Figure 1: Familiarization views of the nano VNA.

A VNA is used to measure S-parameters, which are essentially reflection and transmission coefficients. For a two-port network (which, in our case, are Channel 0 and Channel 1), we have 4 S-parameters: S_{11}, S_{21}, S_{12}, and S_{22}. The subscripts represent the receiving port and the source port, so S_{21} means the signal received at port 2 as a result of a source at port 1 – this is the transmission coefficient from port 1 to port 2.

These nano VNA's can be powered via the built-in battery or the micro-USB cable. To operate the VNA, turn on the device using the on/off slider on the left side of the device. Within a few seconds, you should see a display as shown in Figure 1b. At first, the display will appear cluttered, because you are actually looking at several plots overlayed on one another. The labels for each trace are at the very top of the screen and the default quantities are as follows:

- Yellow: Channel 0's magnitude in log format. This represents the reflection coefficient and is plotted on the rectangular grid. The default scale, as shown, is 10dB per division. Usually, 0dB is at the top of the screen.
- Blue: Channel 1's magnitude in log format. This represents the transmission coefficient and is plotted on the rectangular grid. The default scale, as shown, is 10dB per division. Usually, 0dB is at the top of the screen.
- Magenta: Channel 0's phase in degrees. The default scale is 90°per division.
- Green: This is the smith chart plot of the measured complex impedance of the load (Z_L). It is plotted "behind" the rectangular grid and is on its own scale (normalized **impedance**).

Each scale can be customized, as described in the user's manual, but is not necessary for this lab.

3.2 VNA Setup

To bring up the menu, tap the screen (it helps to use a passive stylus) or press the middle tactile button. Your first task should be to flip the display, which will allow you to turn the ports away from you. To do this, tap the screen, then tap "Display" at the top of the menu. At the bottom, you should then see "Flip Display" – tap this button and the screen should invert. You should also tap the "Pause Sweep" button – this pauses the frequency sweep and measurement cycle of the VNA. Take some time to explore the menu structure – a full map is available in the user's guide.

4 Tasks

4.1 Calibration

When performing RF measurements, you **always** need to calibrate your device to your particular configuration. This calibration needs to be performed every time a cable is connected or the frequency range is changed. First, connect the SMA cables to each port (SMA cables are the smaller coax connectors). On channel 0 (which is confusingly labeled "1" on the circuit board), connect the black SMA-BNC connector, then connect the BNC-grabber clips cable to the open end. On channel 1 (labeled "2" on the circuit board), connect the blue SMA-SMA cable. Don't worry about the different colors or lengths of the cables at this time. RF connectors are sensitive, so when connecting them **DO NOT** overtighten the connectors or cross-thread them. Finger tight is just right in this case.

Before we calibrate the device, you will need to set your frequency range.

1. Tap **Stimulus** from the main menu.

2. Set your frequency range from 50kHz to 5MHz using the **Start** and **Stop** buttons.

You will now follow the calibration procedure using an open, short, and load calibration standard.

1. Every calibration standard will be connected to channel 0. Leave the SMA cable connected to channel 1 disconnected.
2. Bring up the menu and tap **Cal**.
3. Reset the calibration by tapping **Reset**.
4. Leave the grabber clips disconnected from any load. This is an open circuit.
5. Now, under the **Calibration** pane, tap **Open**. The calibration is complete when the background of the text turns black.
6. Connect the grabber clips to form a short circuit.
7. Under the **Calibration** pane, tap **Short**.
8. Once the calibration is complete, move on to the **Load** calibration. In this case, you will connect the grabber clips to a matched load of 50Ω – a simple resistor.
9. Do not conduct a **Thru** measurement. This involves adding another length of transmission line between channel 0 and channel 1.
10. Tap **Done** at the bottom of the **Calibration** pane.
11. If you wish, you may save your calibration by tapping "Save 0". You will be able to recall these calibration settings later.
12. From the **Cal** pane, ensure **Correction** is highlighted in black.
13. Calibration is now complete – tap any portion of the plots to close the menu.

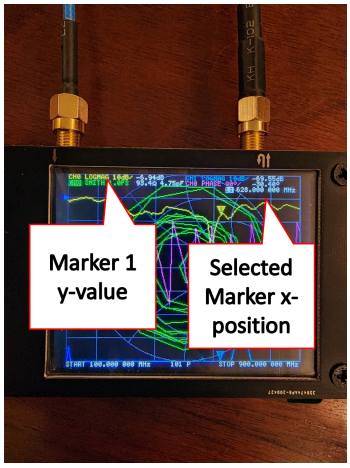


Figure 2: An example readout of a VNA measurement showing the location of Marker position information.

4.2 Measurement

When making measurements, be sure to pause the sweep while changing physical connections. Additionally, you may read values for the marker positions at the top of the screen. The markers may be moved by either tapping/holding and dragging (this requires a stylus) or, choosing **Markers → Select Markers → Marker X** and then using the left/right tactile buttons to move the marker location. Figure 2 shows an annotated example.

1. Given an open circuit load, what type of reflection would you expect to see, in terms of the reflection magnitude and phase? Use the reflection coefficient equation to inform your answer. It may also help to include a screenshot of the resultant plot.

(a) What amplitude and phase do you observe?

Amplitude = Phase =

(b) What position is the open circuit indicating on the Smith Chart? Does this make sense, based on your understanding of the Smith Chart?

2. Given a short circuit load, what type of reflection would you expect to see, in terms of the reflection magnitude and phase? Use the reflection coefficient equation to inform your answer. It may also help to include a screenshot of the resultant plot.

(a) What amplitude and phase do you observe?

Amplitude = Phase =

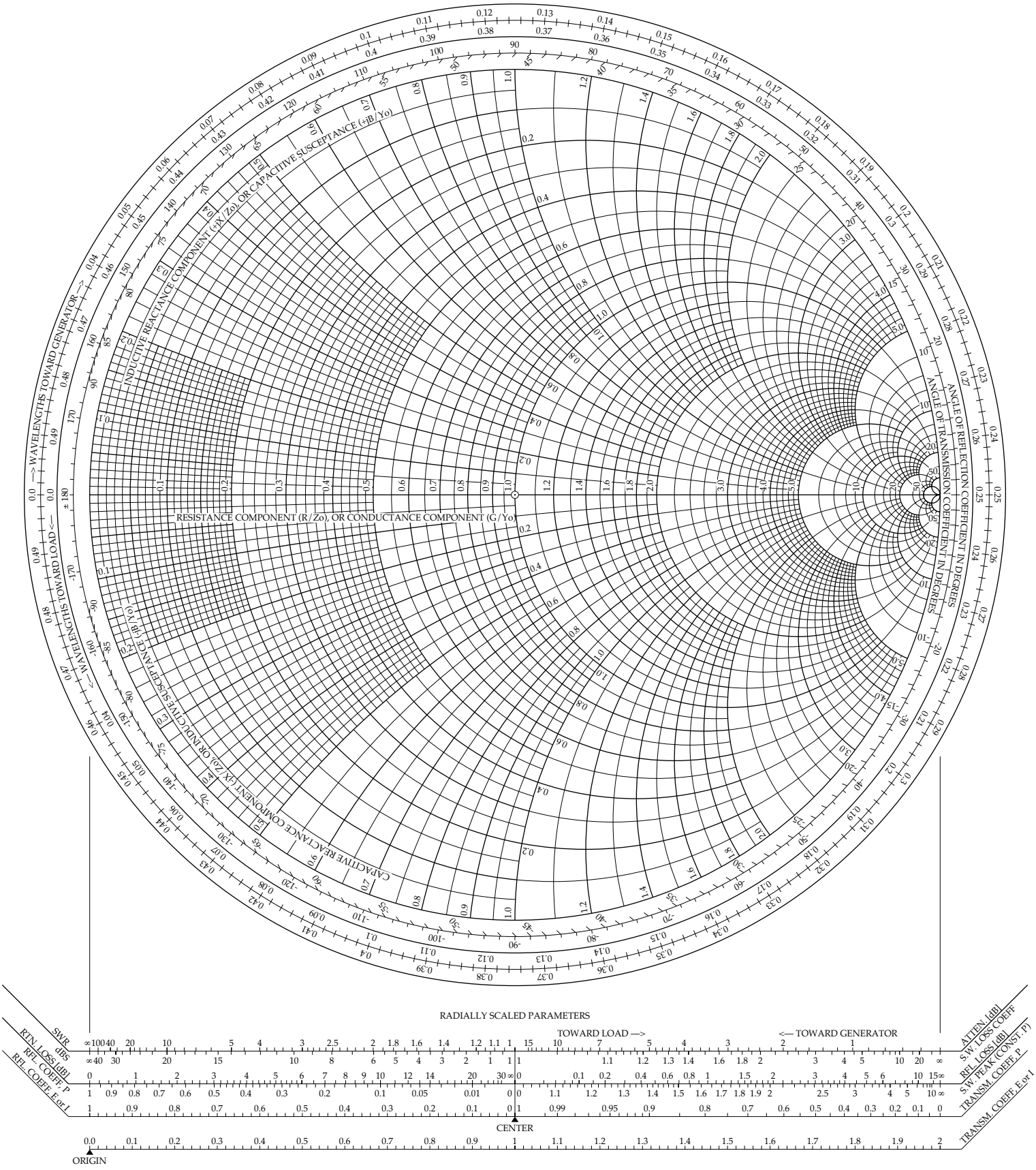
(b) What position is the open circuit indicating on the Smith Chart? Does this make sense, based on your understanding of the Smith Chart?

3. **Imepdance Measurement:** Given a capacitor of value 470 pF, calculate the impedance for the frequency ranges shown in the table below. Plot these values on the smith chart and compare the measurement results – do they agree?

Frequency	**Expected $\mathbf{Z_C}$**
50 kHZ	
100 kHz	
500 kHz	
700 kHz	
900 kHz	
1 MHz	
2 MHz	
3 MHz	
4 MHz	
5 MHz	

The Complete Smith Chart

Black Magic Design



4. **Matched Load Measurement:** Now, you will measure a matched load of 50Ω . Simply connect the grabbers to each side of the same 50Ω resistor used during calibration.

   (a) What is the predicted magnitude of the matched load reflection coefficient, in dB? Remember, you are measuring **Voltages**.

   (b) What is the measured magnitude of the matched load reflection coefficient.

   (c) Identify the resistance value from the marker readout and the Smith Chart.

   (d) Make at least one observation about your measurements.

5. **Mismatched Load Measurement (Resistive):** Now, you will measure two mismatched loads, one purely resistive and one complex. First, connect the 1kΩ resistor to the mini grabbers and answer the following questions.

   (a) What is the predicted magnitude/phase of the reflection coefficient for this system, in dB? Remember, you are measuring **Voltages**.

   (b) What is the measured magnitude/phase of the load reflection coefficient.

   Amplitude = Phase =

   (c) Identify the resistance value from the marker readout and the Smith Chart.

   (d) Make at least one observation about your measurements.

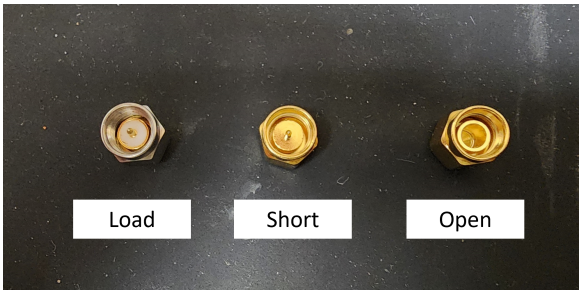


Figure 3: The RF calibration standards for this VNA, used in Part 6.

6. Black Box Measurement

Now, you will measure an unknown device, often referred to as the "Device Under Test" (DUT). The label of the DUT will be covered with tape – do not remove the tape! Also, note the DUT has SMA connectors – disconnect the BNC cable with the minigrabbers. Due to the operational bandwidth of the DUT, you will need to adjust the frequency range to correlate to your specific filter and recalibrate. The calibration procedure is slightly different, in that SMA cables use different calibration standards (as shown in Figure 3. Please use the short, open, and load connectors. This time, you may also perform a thru calibration by connecting channel 0 and channel 1 using a coupler, but this is optional. Now, connect the DUT to the VNA using the appropriate cables and restart the sweep.

(a) Looking at the plot of the transmission and reflection coefficient, what type of filter do you have? How do you know?

(b) Determine the cutoff frequency (or frequencies) of the filter (remember, this is the -3dB point).

(c) Using the Smith Chart display, observe whether the DUT is capacitive or inductive over certain frequency ranges. If you move the Smith Chart marker, you will see the reactance change – can you correlate that change with a value of capacitance or inductance?

(d) Make at least one additional observation about your measurements.